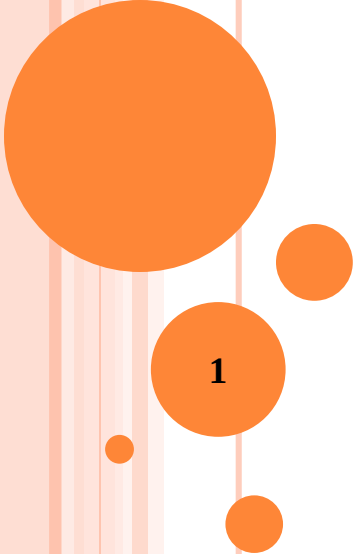


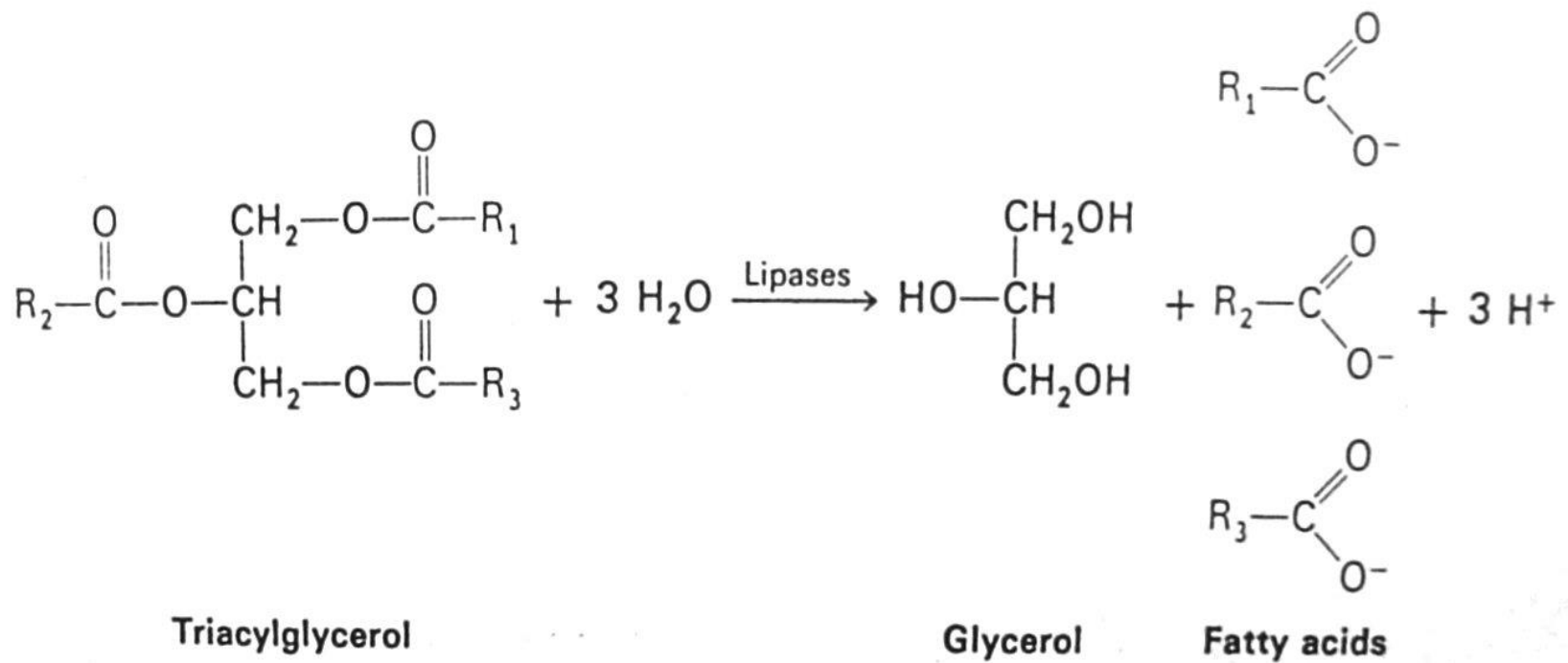
FATTY ACID OXIDATION AND KETONE BODIES

Dr M Kwangu, PhD

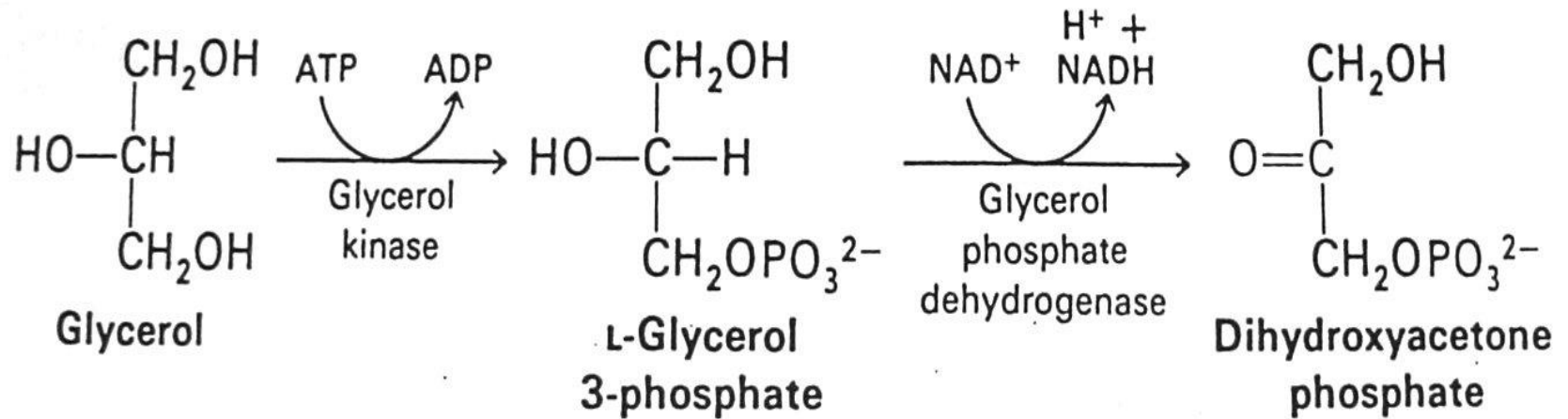


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- The initial event in the utilization of fat as an energy source is the hydrolysis of triacylglycerol by lipases



- Epinephrine, norepinephrine, glucagon, and adrenocorticotrophic hormone stimulate the adenylate cyclase of adipose cells, and thus cause lipolysis.



- It is an aerobic process requiring the presence of oxygen
- Increased fatty acid oxidation is a characteristic of starvation and of diabetes mellitus, leading to ketone body formation by the liver (ketosis)
- Cos gluconeogenesis is dependent on fatty acid oxidation, any impairment of fatty acid oxidation leads to hypoglycaemia

Fatty acids are synthesized and degraded by different mechanisms:

- Fatty acids are both oxidized to acetyl-CoA and synthesized from acetyl-CoA. Although the starting material of one process is identical to the product of the other, fatty acid oxidation is not the simple reverse of fatty acid biosynthesis. It is an entirely different process taking place in separate compartment of the cell. This allows each process to be individually controlled.

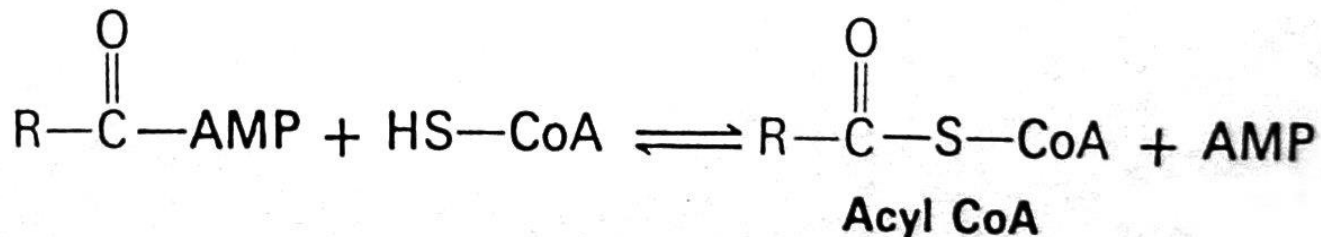
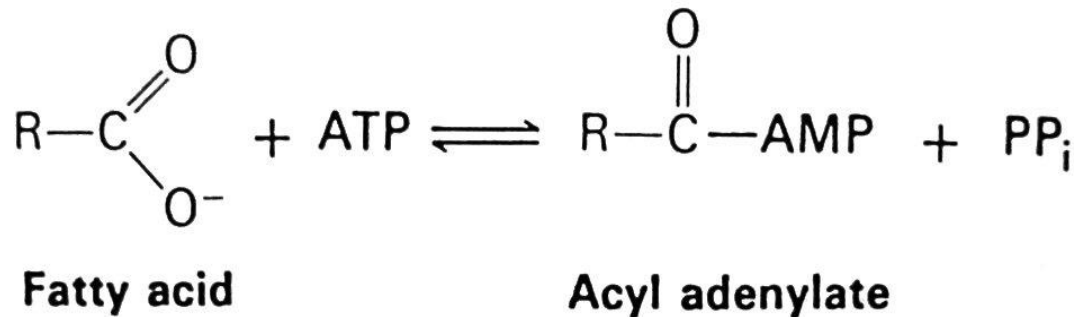
Synthesis

- Cytosol
- Intermediates are covalently linked to ACP
- Fatty acid synthase contain multienzyme activities
- Utilizes NADP⁺ as coenzyme
- Requires both ATP and bicarbonate ion

Oxidation

- Mitochondrial matrix
- Bonded to CoA
- Degradative enzymes are not associated
- Utilizes NAD⁺ and FAD as coenzymes
- Generates ATP
- Aerobic process

STEP 1: ACTIVATION OF FATTY ACIDS TO ACYL-CoA: ACYL-CoA SYNTHETASES ARE FOUND IN THE ENDOPLASMIC RETICULUM, PEROXISOMES, AND INSIDE AND ON THE OUTER MEMBRANE OF MITOCHONDRIA. SEVERAL ACYL-CoA SYNTHETASES HAVE BEEN DESCRIBED, EACH FOR FATTY ACIDS OF DIFFERENT CHAIN LENGTH.



Step 2: Long-chain fatty acids penetrate inner mitochondrial membrane as carnitine derivatives. Carnitine (β -hydroxy- γ -trimethylammonium butyrate) is widely distributed and abundant in muscle. Carnitine palmitoyltransferase-I, present in the outer mitochondrial membrane, catalyzes the following reaction:

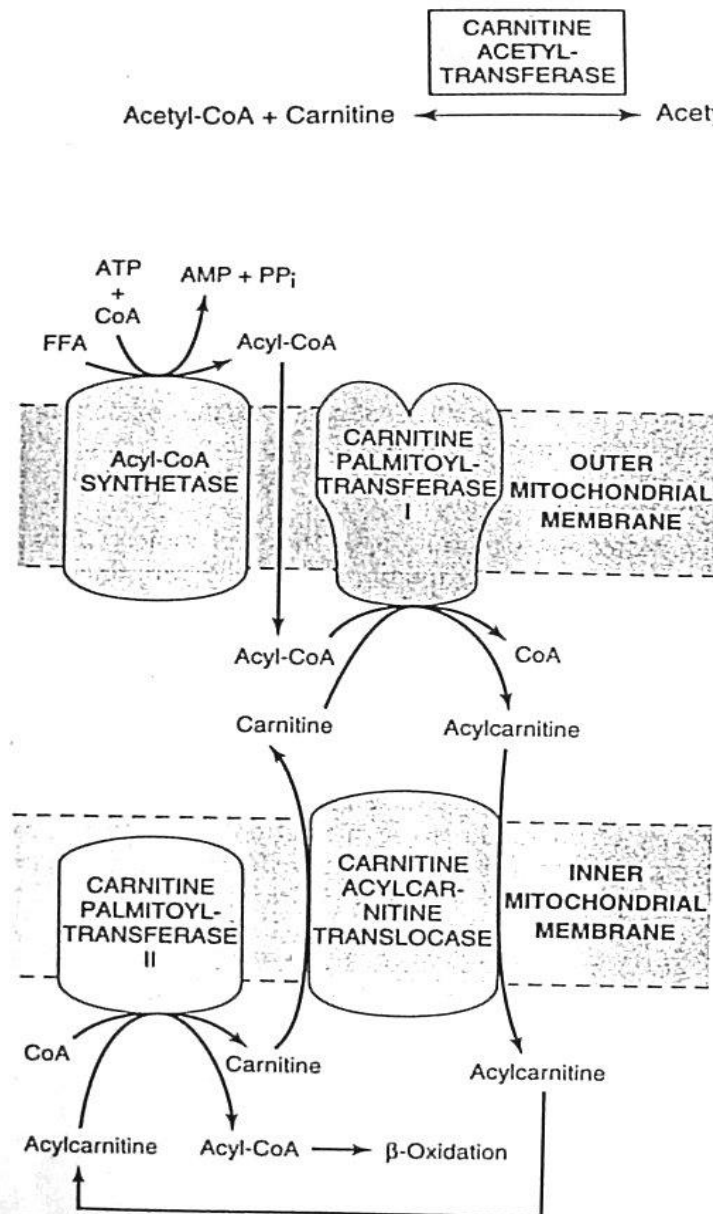


Figure 24–1. Role of carnitine in the transport of long-chain fatty acids through the inner mitochondrial membrane. Long-chain acyl-CoA cannot pass through the inner mitochondrial membrane, but its metabolic product, acylcarnitine, can.

Short and medium-chain fatty acids can cross the mitochondrial membrane by passive diffusion and are activated to their CoA derivatives within the mitochondrion

Very-long-chain fatty acids from the diet are shortened to long-chain fatty acids in the peroxisomes

Long chain fatty acids (16 ± 4 carbons) are the major components of storage triglycerides and dietary fats

Size class	No of carbons	Site of catabolism	Membrane transport
Short-chain	2-4	Mitochondrion	Diffusion
Medium chain	4-12	Mitochondrion	Diffusion
Long-chain	12-20	Mitochondrion	Carnitine cycle
Very long chain	>20	peroxisome	?

The carnitine-mediated entry process is the **rate limiting** step for oxidation of fatty acids in the mitochondria

STEP 3: β -OXIDATION OF FATTY ACIDS INVOLVES SUCCESSIVE CLEAVAGE WITH RELEASE OF ACETYL-CoA. FATTY ACID OXIDASE ARE FOUND IN THE MITOCHONDRIAL MATRIX OR INNER MEMBRANE ADJACENT TO THE RESPIRATORY CHAIN IN THE INNER MEMBRANE.

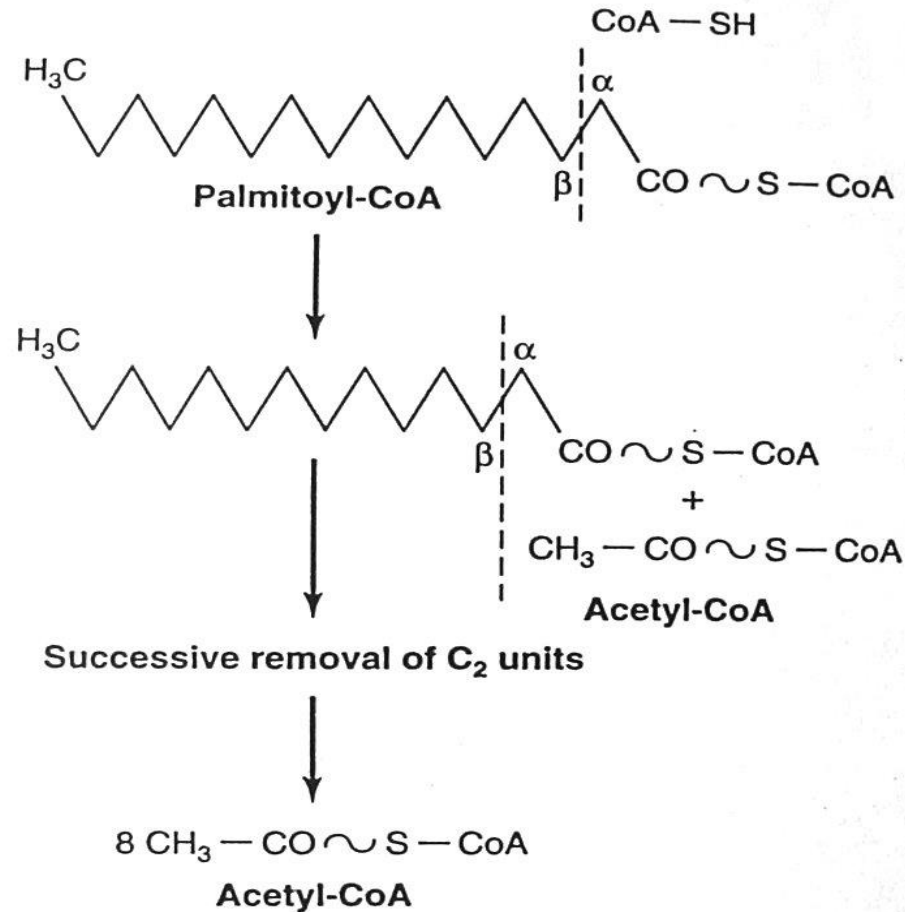


Figure 24–2. Overview of β -oxidation of fatty acids.

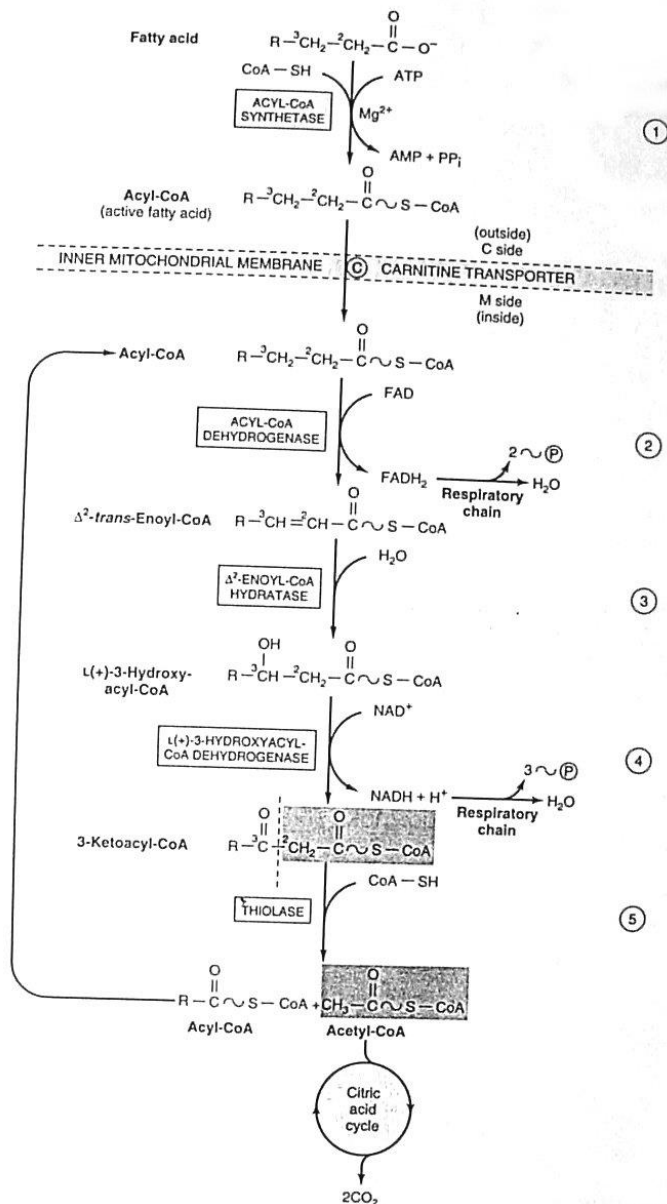


Figure 24-3. β -Oxidation of fatty acids. Long-chain acyl-CoA is cycled through reactions 2-5, acetyl-CoA being split off each cycle, by thiolase (reaction 5). When the acyl radical is only four carbon atoms in length, two acetyl-CoA molecules are formed in reaction 5.

Oxidation of unsaturated fatty acids occurs by a modified β -oxidation pathway:

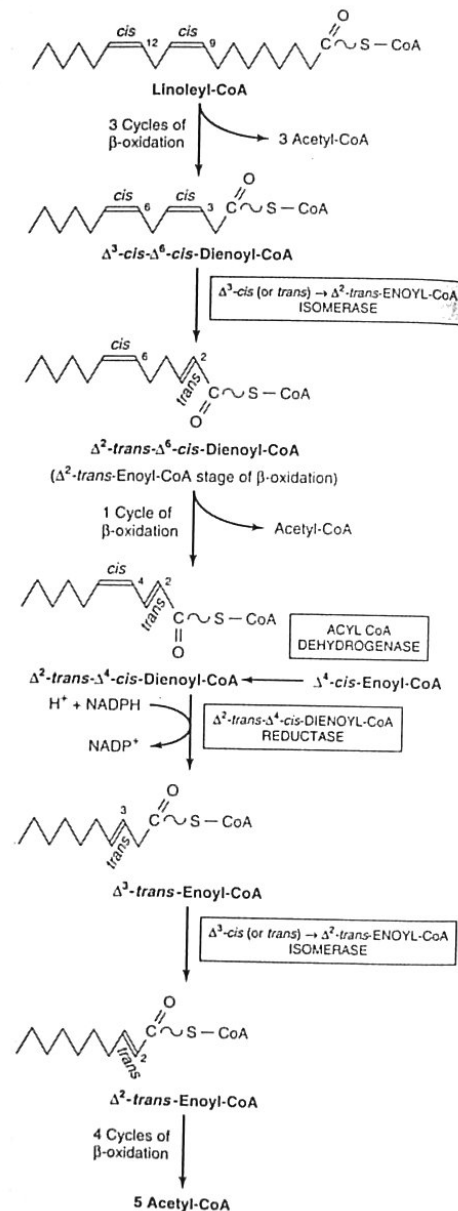


Figure 24-4. Sequence of reactions in the oxidation of unsaturated fatty acids, eg, linoleic acid. Δ^4 -cis-fatty acids or fatty acids forming Δ^4 -cis-enoyl-CoA enter the pathway at the position shown. NADPH for the dienoyl-CoA reductase step is supplied by intramitochondrial sources such as glutamate dehydrogenase, isocitrate dehydrogenase, and NAD(P)H transhydrogenase.

Oxidation of fatty acids produces a large quantity of ATP

Eg Oxidation of one molecule of palmitate- a 16 carbon fatty acid to CO_2 and H_2O (Assuming mitochondrial oxidative phosphorylation produces 2 ATP per FADH_2 oxidized and 3 per NADH oxidized)

Enzyme	Number of NADH or FADH_2	No. of ATP formed
β-Oxidation		
Acyl-CoA dehydrogenase	7 FADH_2	14
β hydroxyacyl-CoA dehydrogenase	7 NADH	21
Citric acid cycle		
Isocitrate dehydrogenase	8 NADH	24
α -ketoglu. dehydrogenase	8 NADH	24
Succinyl-CoA synthetase		8
Succinate dehydrogenase	8 FADH_2	16
Malate dehydrogenase	8 NADH	24
Total		131

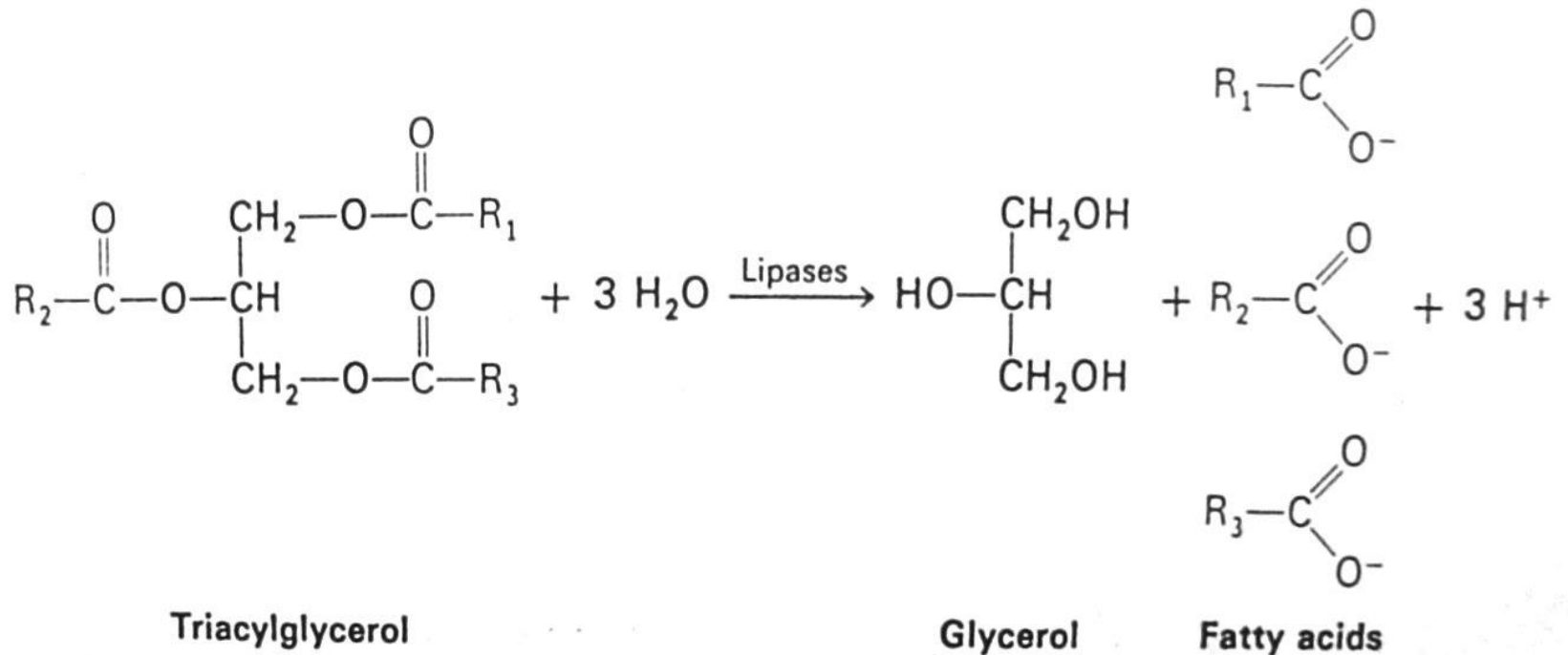
Because activation involves breaking of 2 high energy phosphoanhydride bonds the net gain is 129 ATP

Caloric value of glucose and palmitate

Substrate	Mol. Wt	Net ATP yield (mol/mol)	ATP (mol/g)	Cal value (kJ/g)
Glucose	180	36-38	0.2	17
Palmitate	256	129	0.5	37

- The overall production of ATP per gram of palmitate is about twice that per gram of glucose, because glucose is already partially oxidized compared to palmitate
- The caloric value of fats is about twice that of sugars.

- The initial event in the utilization of fat as an energy source is the hydrolysis of triacylglycerol by lipases.



TRIACYLGLYCEROLS ARE HIGHLY CONCENTRATED ENERGY STORES

- Reduced and anhydrous; Complete oxidation of fatty acids yields 9 kCal/g, whereas, proteins and carbohydrates yield 4 kCal/g.

A 70 kg man:

100,000 kCal in triacylglycerols
25,000 kCal in proteins (muscles)
600 kCal in glycogen
400 kCal in glucose

- Triacylglycerols constitute about 11 kg of his total body weight. If this amount were stored in glycogen, his total body weight would be 55 kg greater.
- In mammals, the major site of accumulation of triacylglycerols is the cytoplasm of adipose cells (fat cells). Droplets of triacylglycerol coalesce to form a large globule, which may occupy most of the cell volume.
- Adipose cells are specialized for the synthesis and storage of triacylglycerols and for their mobilization into fuel molecules that are transported to other tissues by the blood.

PEROXISOMES OXIDIZE VERY LONG CHAIN FATTY ACIDS:

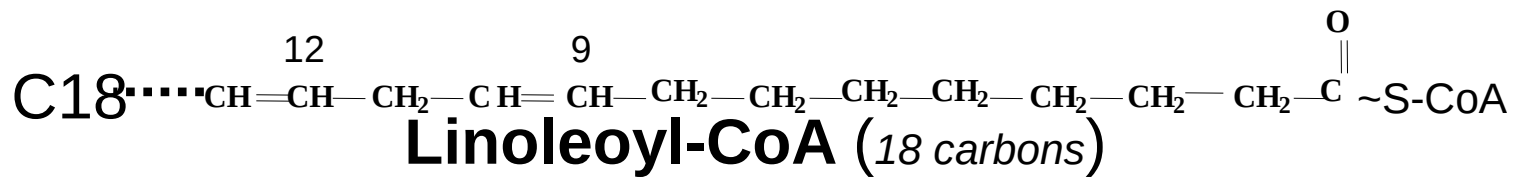
- Very long chain acyl-CoA synthetase facilitates the oxidation of very long chain fatty acids (e.g., C20, C22). These enzymes are induced by high-fat diets and by hypolipidemic drugs such as Clofibrate. β -oxidation takes place and ends at octanoyl-CoA.
- It is subsequently removed from the peroxisomes in the form of octanoyl and acetylcarnitine, and both are further oxidized in mitochondria.

a- and w-oxidation of fatty acids are specialized pathways

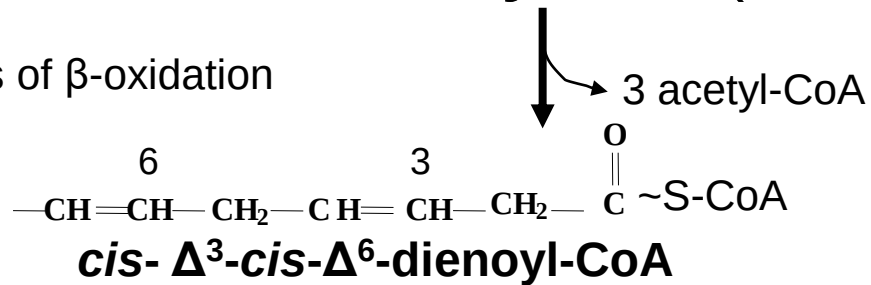
- a-oxidation i.e., removal of one carbon at a time from the carboxyl end of the molecule has been detected in brain tissue. It does not generate CoA intermediates and does not generate high-energy phosphates.
- w-oxidation is a minor pathway and is brought about by cytochrome P450 in the endoplasmic reticulum. CH_3 group is converted to a $-\text{CH}_2\text{OH}$ group that subsequently is oxidized to $-\text{COOH}$, thus forming a dicarboxylic acid. They subsequently undergo β -oxidation and are excreted in the urine.

OXIDATION OF UNSATURATED FATTY ACIDS OCCURS BY A MODIFIED β -OXIDATION

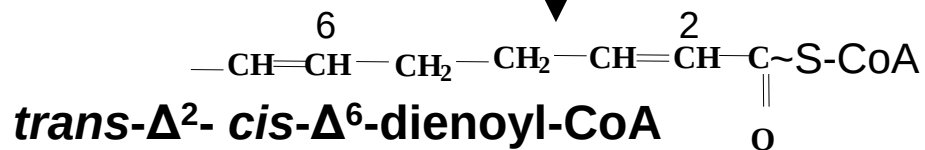
- Unsaturated fatty acids are already partially oxidized, so less FADH₂ and correspondingly less ATP is produced by their oxidation
- The double bonds in polyunsaturated fatty acids have *cis* geometry and occur at three carbon intervals, where as the intermediates in β -oxidation have *trans* geometry and the reactions proceed in two-carbon steps
- The metabolism of unsaturated fatty acids require additional enzymes to shift the position and change the geometry of the double bonds
- The CoA esters of these acids are degraded by the enzymes normally responsible for the β -oxidation until either a *cis*- Δ^3 -acyl-CoA or a *cis*- Δ^4 -acyl-CoA compound is formed depending upon the position of the double bonds
- The Δ^3 -*cis*-acyl-CoA is isomerised to the corresponding Δ^2 -*trans*-CoA stage of β -oxidation for subsequent hydration and dehydration
- Any Δ^4 -*cis*-acyl-CoA either remaining or entering the pathway at this point after conversion by *acyl-CoA dehydrogenase* to *trans*- Δ^2 - *cis*- Δ^4 -dienoyl-CoA is reduced to *trans*- Δ^3 - enoyl-CoA, which is isomerised to *trans*- Δ^2 -Enoyl-CoA
- Odd-numbered double bonds are handled by *isomerase* and even-numbered by *isomerase* and *reductase*



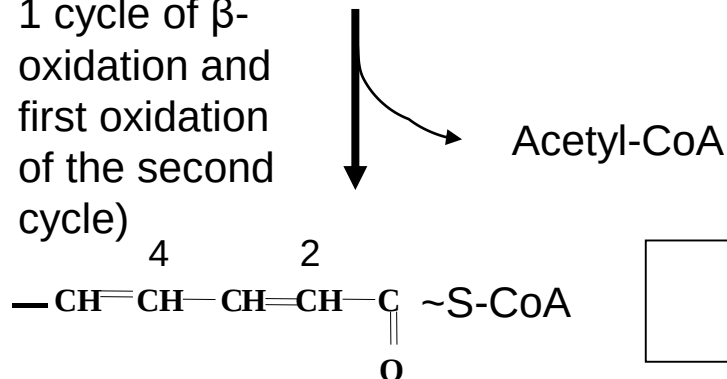
3 cycles of β -oxidation



cis - (or *trans*) $\Delta^3 \rightarrow$ *trans*- Δ^2 -Enoyl-CoA
isomerase



1 cycle of β -oxidation and first oxidation of the second cycle)

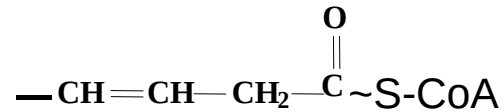


Acyl-CoA dehydrogenase

$\text{H}^+ + \text{NADPH}$

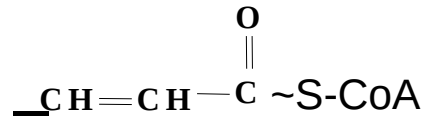
NADP^+

trans- Δ^2 -cis- Δ^4 -dienoyl-CoA
reductase



trans- Δ^3 -Enoyl-CoA

*cis (or trans) - Δ^3 - $\longrightarrow$ *trans- Δ^2 -Enoyl-CoA*
*isomerase**



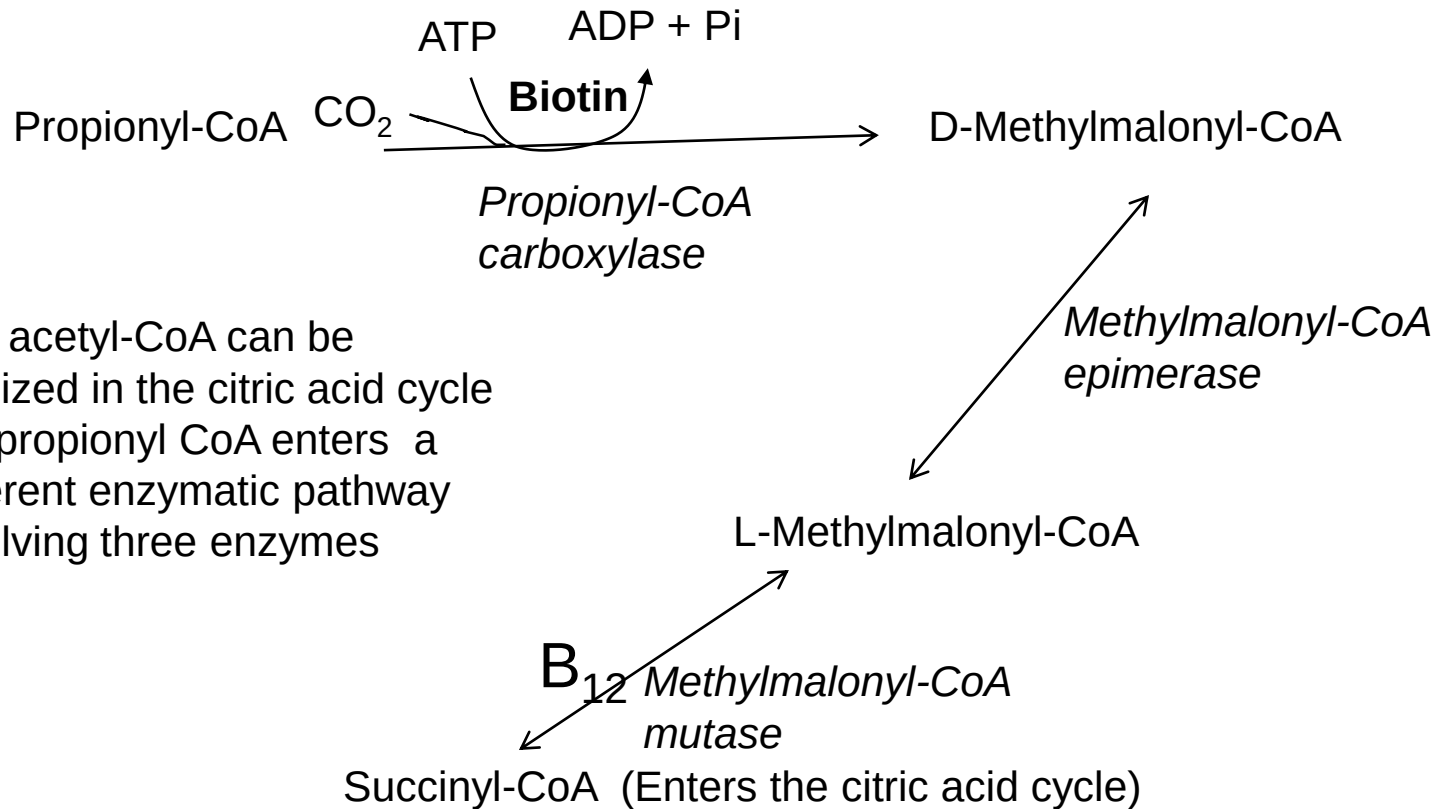
trans- Δ^2 -Enoyl-CoA

Four cycles of
oxidation

Five Acetyl-CoA

Oxidation of fatty acids with an odd number of carbon atoms yields acetyl-CoA plus propionyl-CoA

- Initially oxidation occurs as in even –numbered acids beginning at the carboxyl end
- The substrate for the last cycle is a five-carbon acyl-CoA
- When this is cleaved the products are acetyl-CoA and propionyl-CoA



The acetyl-CoA can be oxidized in the citric acid cycle but propionyl CoA enters a different enzymatic pathway involving three enzymes

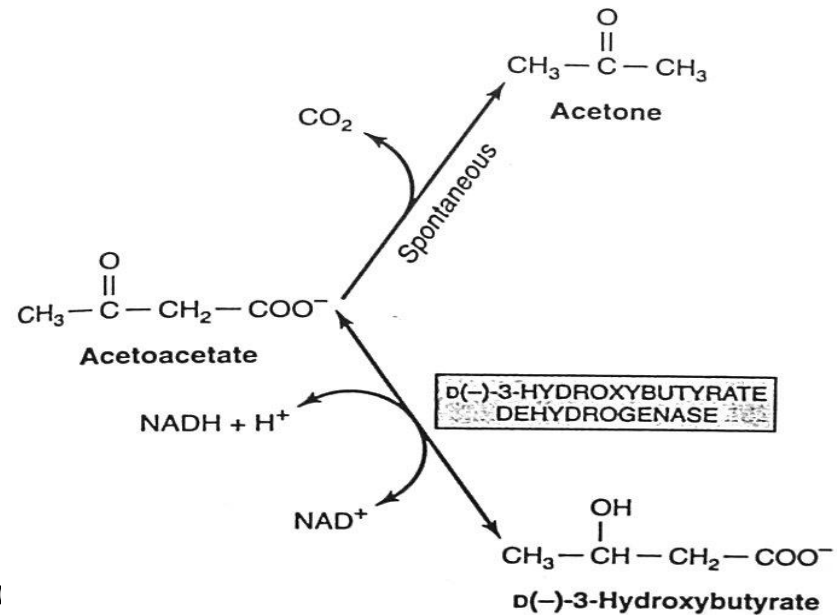
- The propionyl residue from an odd-chain of fatty acid is the only part of a fatty acid that is glucogenic

Principal reactions in fatty acid oxidation

Step	Reaction	Enzyme
1	$\text{Fatty acid} + \text{CoA} + \text{ATP} \rightleftharpoons \text{acyl CoA} + \text{AMP} + \text{PP}_i$	Acyl CoA synthetase (also called fatty acid thiokinase and fatty acid: CoA ligase [AMP])
2	$\text{Carnitine} + \text{acyl CoA} \rightleftharpoons \text{acyl carnitine} + \text{CoA}$	Carnitine acyltransferase
3	$\text{Acyl CoA} + \text{E-FAD} \longrightarrow \text{trans-}\Delta^2\text{-enoyl CoA} + \text{E-FADH}_2$	Acyl CoA dehydrogenases (several enzymes having different chain-length specificity)
4	$\text{trans-}\Delta^2\text{-Enoyl CoA} + \text{H}_2\text{O} \rightleftharpoons \text{L-3-hydroxyacyl CoA}$	Enoyl CoA hydratase (also called crotonase or 3-hydroxyacyl CoA hydrolyase)
5	$\text{L-3-Hydroxyacyl CoA} + \text{NAD}^+ \rightleftharpoons \text{3-ketoacyl CoA} + \text{NADH} + \text{H}^+$	L-3-Hydroxyacyl CoA dehydrogenase
6	$\text{3-Ketoacyl CoA} + \text{CoA} \rightleftharpoons \text{acetyl CoA} + \text{acyl CoA (shortened by C}_2\text{)}$	β -Ketothiolase (also called thiolase)

KETOGENESIS

- It occurs when there is a high rate of fatty acid oxidation in the liver



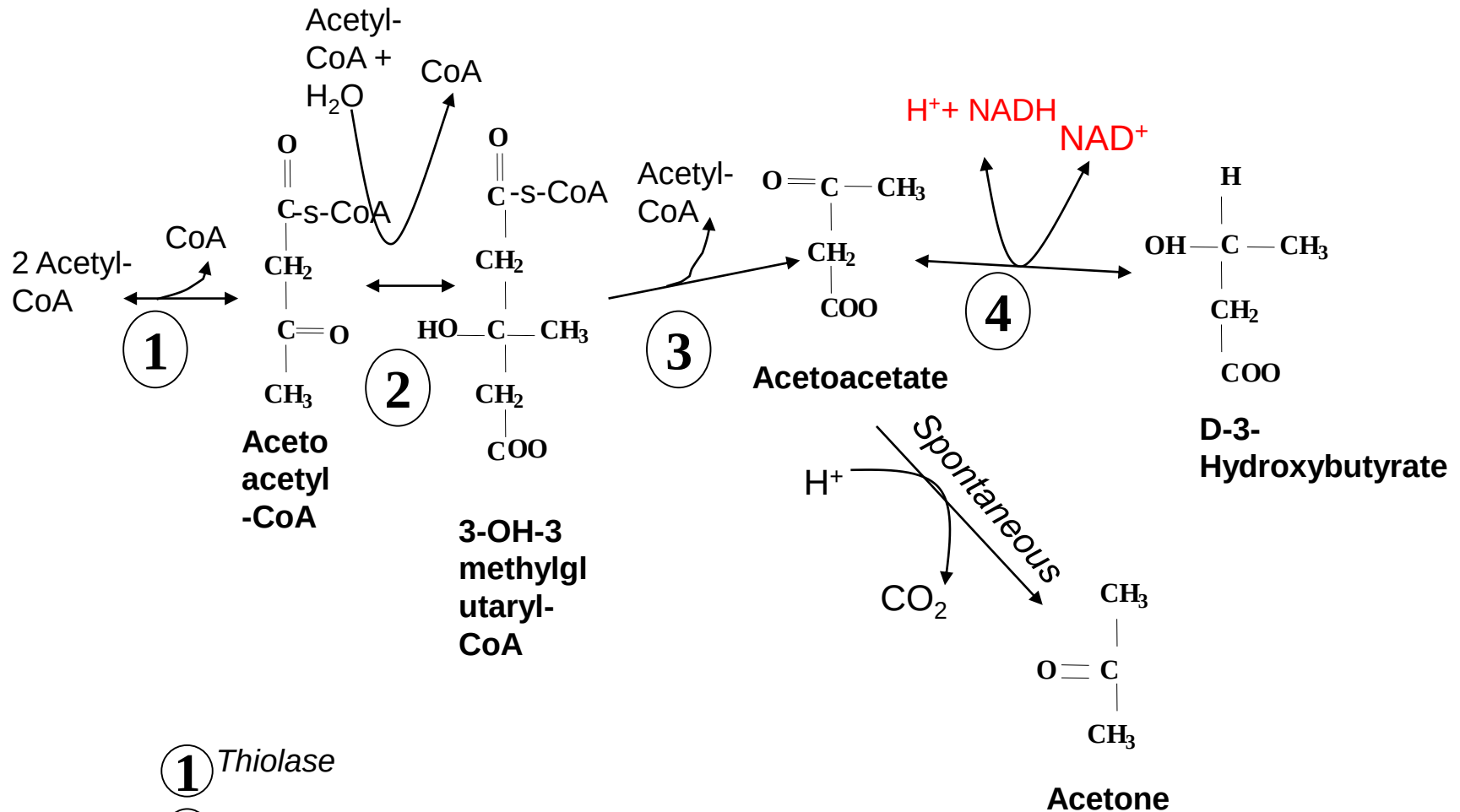
- These three substances are collectively called acetone bodies or ketone bodies. Their formation are associated with mitochondrial fatty acid oxidation.

Figure 24–5. Interrelationships of the ketone bodies. D(-)-3-Hydroxybutyrate dehydrogenase is a mitochondrial enzyme.

Ketone bodies are formed from acetyl-CoA when fat breakdown dominates

- Acetyl-CoA formed from fatty oxidation enters the CAC only if fat and carbohydrate degradation are approximately balanced
 - cos the entry of acetyl-CoA into the CAC depends on the availability of oxaloacetate for the formation of citrate- but concentration of oxaloacetate is lowered if carbohydrate is unavailable or improperly utilized -oxaloacetate is normally formed from pyruvate, the product of glycolysis.
- In fasting or diabetics oxaloacetate is consumed to form glucose by the gluconeogenic pathway and hence unavailable for condensation with acetyl-CoA
- Under these conditions acetyl-CoA is diverted to the formation of acetoacetate, D-3-hydroxybutyrate and acetone - the ketone bodies (occurs solely in the liver)
- The production and export of ketone bodies from the liver to extrahepatic tissues allows continued oxidation of fatty acids in the liver when acetyl-CoA is not being oxidized in the citric acid cycle
- Enzymes responsible for ketone body formation are associated mainly with the mitochondria
- Abnormally high levels of ketone bodies are present in the blood of untreated diabetics.
- D-3-hydroxybutyrate is quantitatively the predominant ketone body in the blood and urine in ketosis

Acetoacetate is formed from acetyl-CoA in three steps



- ① Thiolase
- ② 3-hydroxy-3-methylglutaryl-CoA synthase
- ③ 3-hydroxy-3-methylglutaryl-CoA lyase
- ④ D-3-hydroxybutyrate dehydrogenase

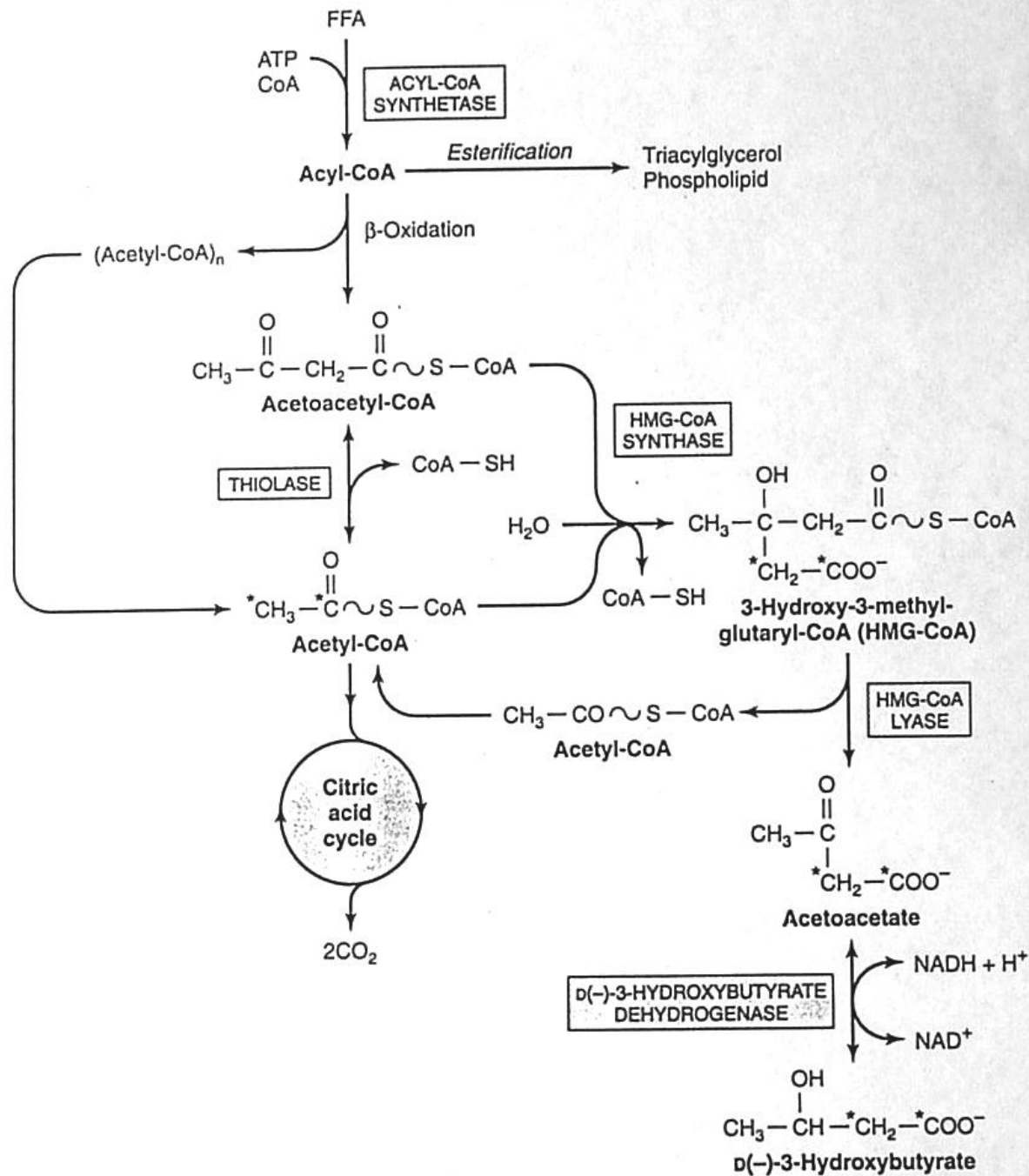


Figure 24-7 Pathways of ketogenesis in the liver (FFA, free fatty acids; HMG, 3-hydroxy-3-methylglutaryl.)

Ketone bodies serve as fuel for extrahepatic tissues

- In the liver acetoacetate cannot be reactivated directly, except in the cytosol where its used in a less active pathway as a precursor in cholesterol synthesis- hence the net production of ketone bodies by the liver
- In extrahepatic tissues, acetoacetate, is activated to acetoacetyl-CoA by *succinyl-CoA acetoacetate-CoA transferase* (NB liver lacks this enzyme)
 - CoA is transferred from succinyl-CoA to form acetoacetyl-CoA
 - The acetoacetyl-CoA is split to acetyl-CoA by *thiolase* and oxidized in the CAC
 - When the concentration of ketone bodies saturate the oxidation machinery, a large proportion of oxygen consumption may be accounted for by the oxidation of ketone bodies
- While acetoacetate and D-3-hydroxybutyrate are readily oxidized in the extrahepatic tissues, acetone is difficult to oxidize in vivo and is to a large extent volatilized in the lungs
- Heart muscle and renal cortex tend to use acetoacetate for energy in preference to glucose

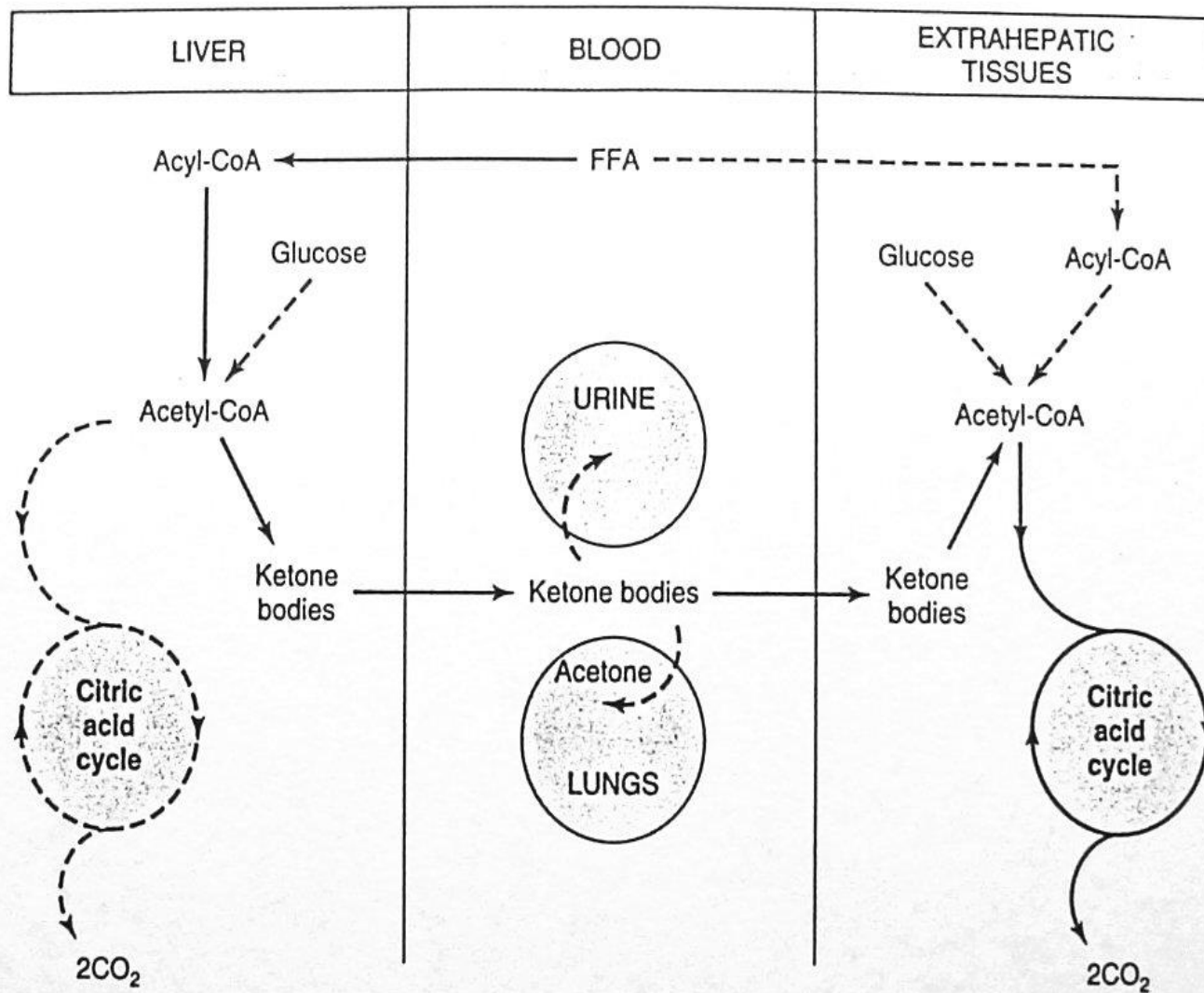


Figure 24-6. Formation, utilization, and excretion of ketone bodies. (The main pathway is indicated by the solid arrows.)

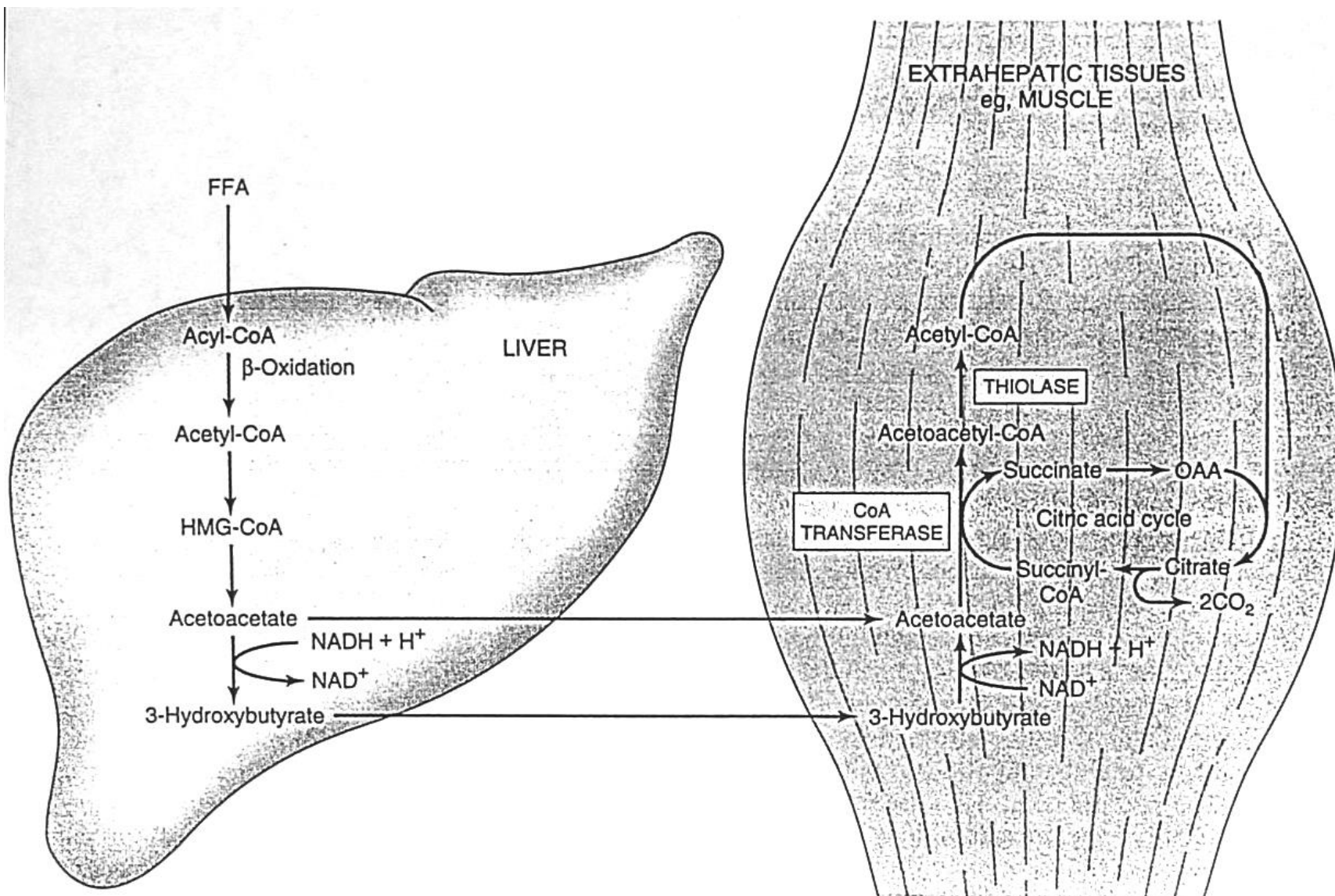
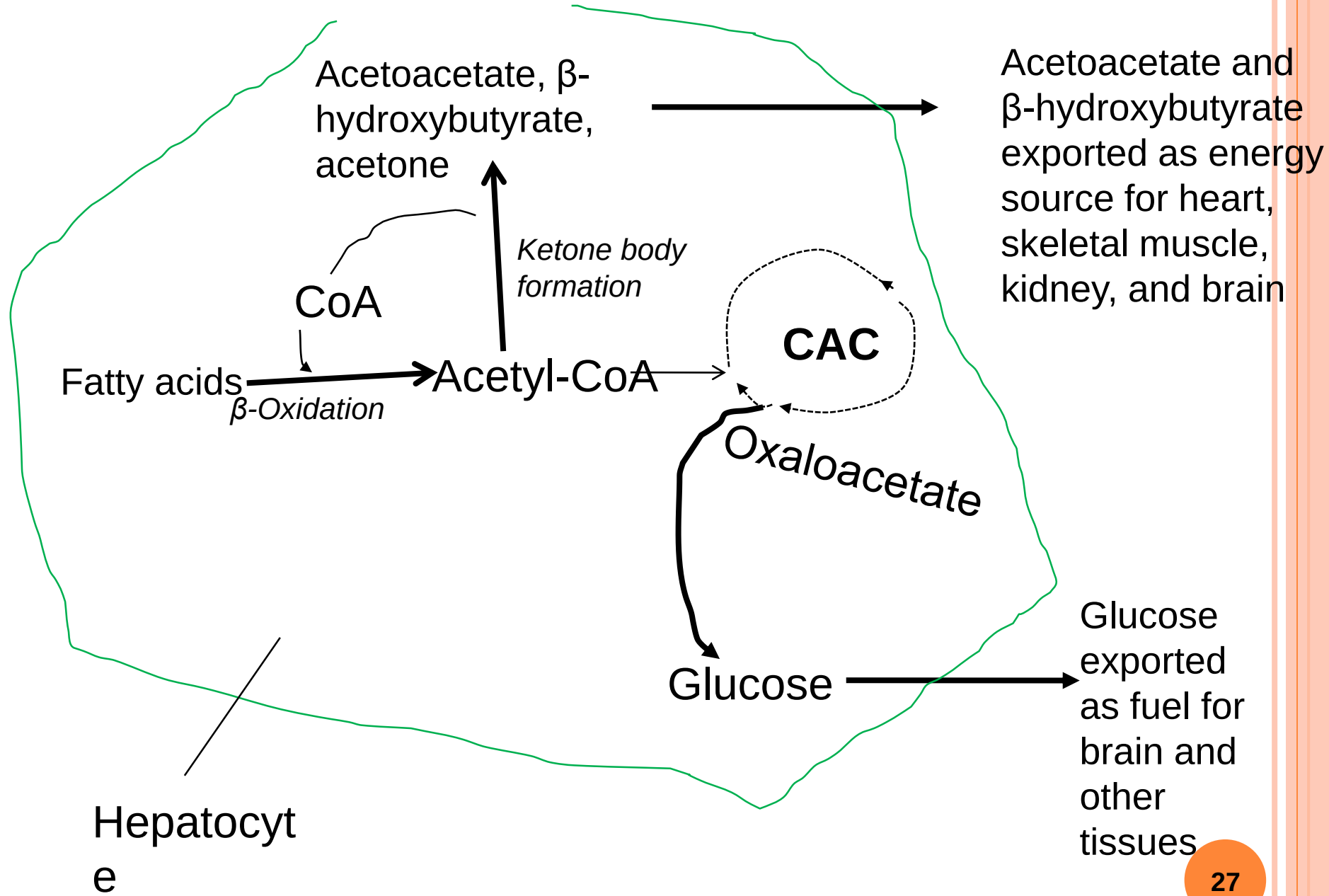


Figure 24–8. Transport of ketone bodies from the liver and pathways of utilization and oxidation in extrahepatic tissues.

Ketone bodies are over produced in diabetes and during starvation

- The production and export of ketone bodies from the liver to extrahepatic tissues allows continued oxidation of fatty acids in the liver when acetyl-CoA is not being oxidized in the citric acid cycle
- When the intermediates of the citric acid cycle are being used for glucose synthesis by gluconeogenesis, oxidation of citric acid cycle intermediates slows, and so does acetyl-CoA oxidation
 - Moreover the liver has a limited supply of CoA, and when most of it is tied up in acetyl-CoA β -oxidation slows
 - The production and export of ketone bodies frees CoA, allowing continued fatty acid oxidation



KETOGENESIS IS REGULATED AT THREE CRUCIAL STEPS:

1. **Adipose tissue:** Factors regulating mobilization of free fatty acids from adipose tissue are important in controlling ketogenesis
2. **Liver:** After acylation, fatty acids undergo β -oxidation or esterified to triacylglycerol or ketone bodies.

a. CPT-1 regulates entry of long-chain acyl groups into mitochondria prior to β -oxidation. Its activity is low in the fed state, and high in starvation.

Fed state: Malonyl-CoA formed in the fed state is a potent inhibitor of CPT-1. Under these conditions, free fatty acids enter the liver cell in low concentrations and are nearly all esterified to acylglycerols and transported out as VLDL.

Starvation: Free fatty acid concentration increases with starvation, acetyl-CoA carboxylase is inhibited and malonyl-CoA decreases releasing the inhibition of CPT-I and allowing more β -oxidation.

These events are reinforced in starvation by decrease in insulin/glucagon ratio. This causes inhibition of acetyl-CoA carboxylase in the liver by phosphorylation.

In short, β -oxidation from free fatty acids is controlled by the CPT-I gateway into the mitochondria, and the balance of free fatty acid uptake not oxidized is esterified.

3. **Acetyl-CoA formed from β -oxidation of fatty acids is either oxidized in TCA cycle or it forms ketone bodies.**

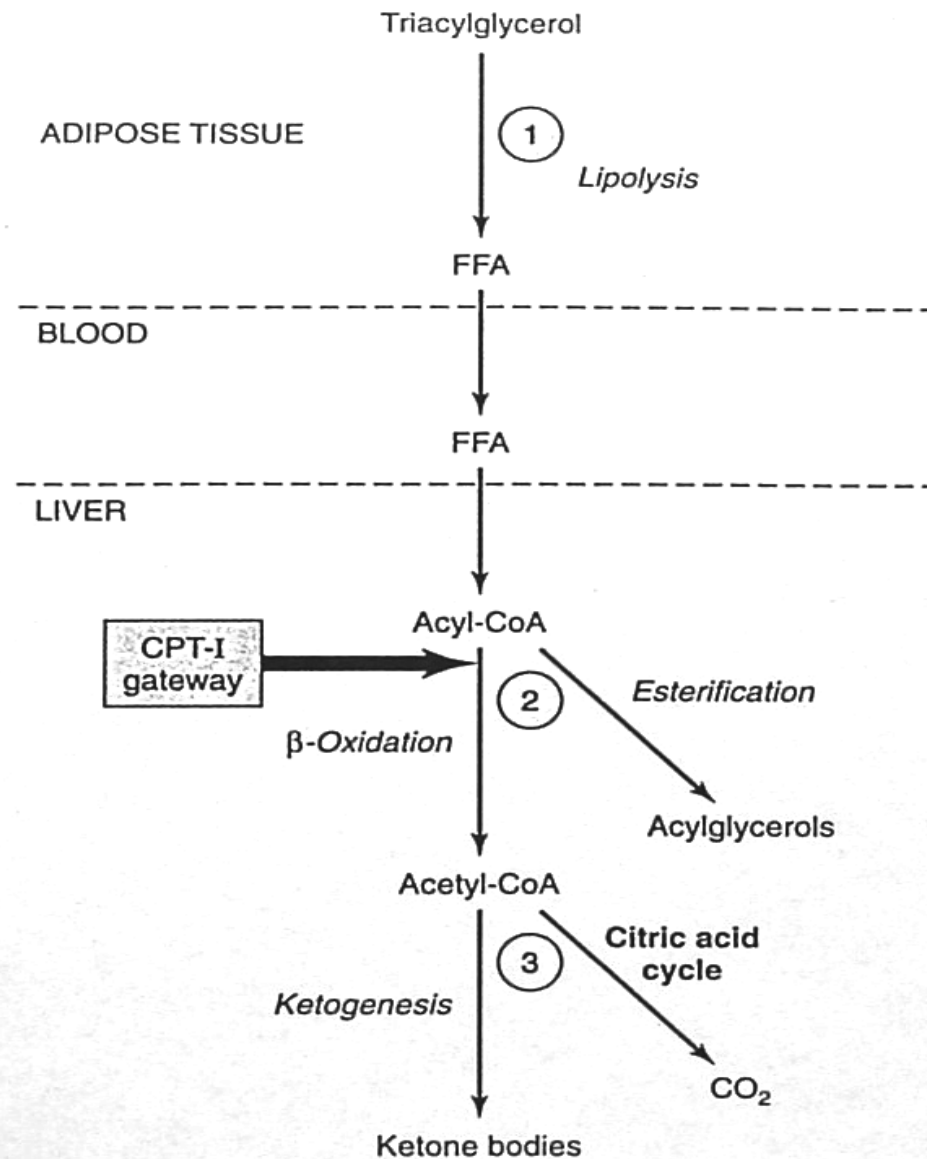


Figure 24–9. Regulation of ketogenesis. ①–③ show three crucial steps in the pathway of metabolism of free fatty acids (FFA) that determine the magnitude of ketogenesis. (CPT-I, carnitine palmitoyltransferase-I.)

- In untreated diabetics, extrahepatic tissues can not take up glucose efficiently from the blood to use as a fuel or to convert to fats for storage
 - Under these circumstances malonyl-CoA (the starting material for fatty acid synthesis) is not formed and thus not present to inhibit carnitine acyltransferase I
 - Fatty acids enter the mitochondria to be degraded to acetyl-CoA, which can not pass through the citric acid cycle as intermediates have been drawn off for gluconeogenesis
 - Accumulation of acetyl-CoA accelerates the formation of ketone bodies beyond the capacity of extrahepatic tissues to oxidize them
 - The increased blood levels of acetoacetate and β -hydroxybutyrate lower the blood pH causing acidosis

Regulation of fatty acid oxidation

- In the liver the fatty acyl-CoA formed in the cytosol has two major pathways open to it
 - β -oxidation by enzymes in the mitochondria
 - Conversion into triacylglycerols and phospholipids by enzymes in the cytosol
- The pathway taken depends on the rate of transfer of long-chain fatty acyl-CoA into the mitochondria
- The transfer of fatty acyl groups into the mitochondrial matrix is rate-limiting for β -oxidation. Once inside the mitochondria the fatty acyl groups are committed to oxidation to acetyl-CoA

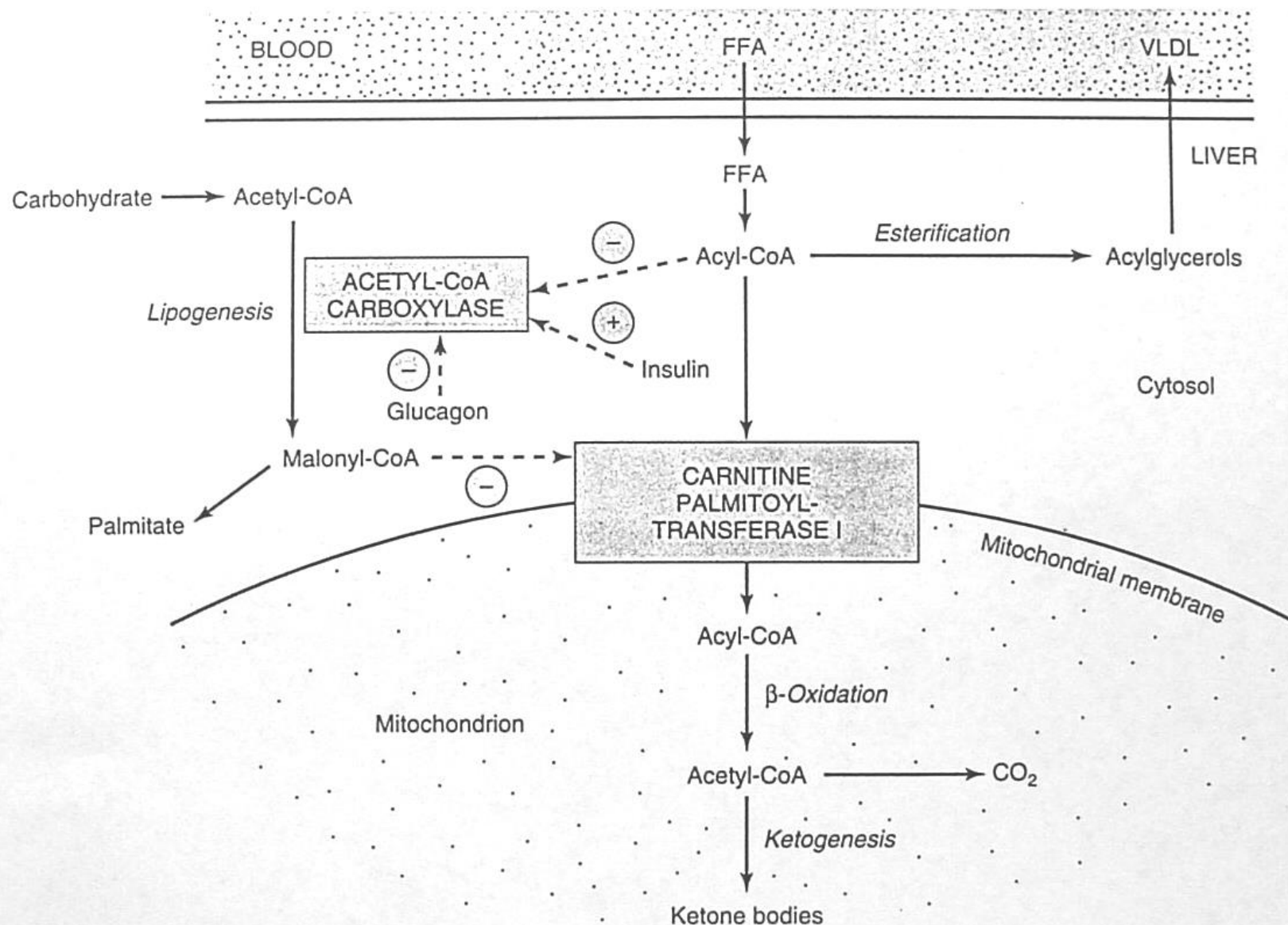


Figure 24–10. Regulation of long-chain fatty acid oxidation in the liver. (FFA, free fatty acids; VLDL, very low density lipoprotein.) Positive ($\oplus$) and negative ($\ominus$) regulatory effects are represented by broken arrows and substrate flow by

- Malonyl- CoA, the first intermediate in the cytosolic biosynthesis of long-chain fatty acids from acetyl-CoA increases in concentration when an animal is well supplied with carbohydrate.
- Excess glucose that can not be oxidized or stored as glycogen is converted in the cytosol into fatty acids for storage as triacylglycerol
- Malonyl- CoA inhibits carnitine acyltransferase I
- This ensures that fatty acid oxidation is inhibited when the liver is amply supplied with glucose as fuel and actively synthesizing triacylglycerols from excess glucose
- In addition, a high $[NADH]/[NAD^+]$ ratio inhibits β -hydroxyacyl-CoA dehydrogenase
- High concentrations of acetyl-CoA inhibit thiolase

CLINICAL ASPECTS:

- **Carnitine-deficiency:** can occur in newborn or preterm infants owing to inadequate biosynthesis or renal leakage. Losses can also occur in hemodialysis.
 - Symptoms: hypoglycemia due to reduced gluconeogenesis resulting from impaired fatty acid oxidation, resulting in muscle weakness (Reye's syndrome).
 - Carnitine is supplemented with diet.
- **Deficiency of Carnitine palmitoyltransferase-I and -II:**
 - I Deficiency– affects only liver, resulting in reduced fatty acid oxidation and ketogenesis with hypoglycemia.
 - II Deficiency– skeletal muscle
 - Sulfonylureas (glyburide and tolbutamide) inhibit CPT and reduce fatty acid oxidation
- **Inherited defects in the β -oxidation lead to non-ketotic hypoglycemia, coma, and fatty liver.**
- **Jamaican vomiting sickness:** It is caused by eating unripe fruit of the akee tree which contains a toxin, hypoglycin, that inactivates medium-and short-chain acyl-CoA dehydrogenase, inhibiting β -oxidation resulting in hypoglycemia with excretion of medium- and short-chain mono- and dicarboxylic acids.

○ **Ketoacidosis results from prolonged ketosis:**

- **Ketonemia-** higher than normal quantities of ketone bodies in blood
- **Ketonuria-** higher than normal quantities of ketone bodies in urine.
- **Ketosis:** the overall condition is called ketosis.